

A Wide Power Dynamic Range High Efficiency Triple-path Charge Pump System for RF Energy Harvesting Applications

Zeyu Yang

School of Electronics and
Communication Engineering
Sun Yat-sen University
Shenzhen, China
yangzy27@mail2.sysu.edu.cn

Yichen Zhang

School of Electronics and
Communication Engineering
Sun Yat-sen University
Shenzhen, China
zhangych77@mail2.sysu.edu.cn

Weijie Ge

School of Electronics and
Communication Engineering
Sun Yat-sen University
Shenzhen, China
gewj@mail2.sysu.edu.cn

Jian Guan

School of Electronics and
Communication Engineering
Sun Yat-sen University
Shenzhen, China
guanj27@mail2.sysu.edu.cn

Xinying Su

School of Electronics and
Communication Engineering
Sun Yat-sen University
Shenzhen, China
suxy55@mail.sysu.edu.cn

Jingjing Liu*

School of Electronics and
Communication Engineering
Sun Yat-sen University
Shenzhen, China
liujj77@mail.sysu.edu.cn

Abstract—This paper proposes an ultra-wide power dynamic range (PDR) charge pump system for RF energy harvesting, operating at 2.45 GHz, with three selectable operating modes. Two distinct RF-DC rectifiers are designed to deal with low-power, mid-power and high-power conditions, and a basic control circuit is used to select the operating mode, thereby expanding the input power range over which high power conversion efficiency (PCE) is also maintained. The proposed charge pump system was simulated using a standard 0.18 μm CMOS process, achieving a sensitivity of -21 dBm and a peak PCE of 61.4% at -10 dBm. When operating at 2.45 GHz with a 100 k Ω load, the proposed charge pump maintains a PCE above 20% over an input power range of -22 dBm to 7 dBm.

Keywords—energy harvesting, charge pump, dynamic range

I. INTRODUCTION

With the rapid development of the Internet of Things (IoT) and Radio Frequency Identification (RFID) technologies, as well as the widespread application of wearable devices and implantable medical devices, the limitations of traditional battery powered methods are becoming increasingly prominent [1]-[3]. IoT devices generally face issues such as limited battery life, high maintenance costs, and poor environmental adaptability. Especially in harsh deployment scenarios such as industrial monitoring and deep-sea exploration, regular battery replacement is almost impossible to achieve. This practical demand has driven a research boom in radio frequency energy harvesting (RFEH) technology, which captures electromagnetic wave energy in the environment and converts it into usable direct current energy, providing the possibility of achieving permanent self-powered systems. Fig. 1 shows the concept of a RFEH system. In this system, the RF energy source originates from ubiquitous electromagnetic waves. Antenna captures

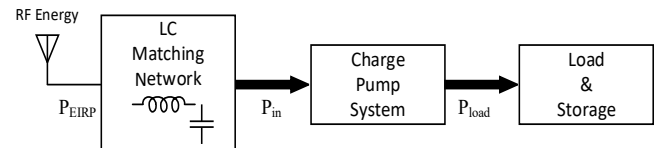


Fig. 1. Concept of a RFEH system.

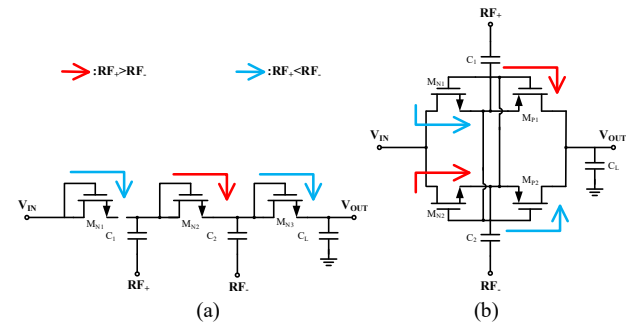


Fig. 2. (a) The conventional Dickson charge pump, (b) the conventional CC charge pump.

incident electromagnetic waves, and through matching network, feeds the harvested energy into the harvesting circuit. Subsequently, the rectifier then converts the RF power into DC power and supply to energy storage devices and loads. During the energy harvesting process, energy gradually decreases from the power captured by the antenna (Effective Isotropic Radiated Power, P_{EIRP}) to the input power delivered to the charge pump system (P_{in}) and finally to the power obtained by the load (P_{load}). This phenomenon renders the PCE a critical performance indicator of the entire system. [4]-[7].

As the core component of the energy harvesting system, the PCE of the RF-DC rectifier, which is charge pump here, directly affects the PCE of the entire system. Fig. 2 presents the circuit structures and operating principles of the conventional Dickson

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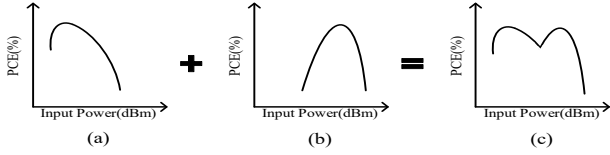


Fig. 3. Conventional rectifier's performance efficiencies (a) at low power, (b) at high power, and (c) combination of the two power scenarios.

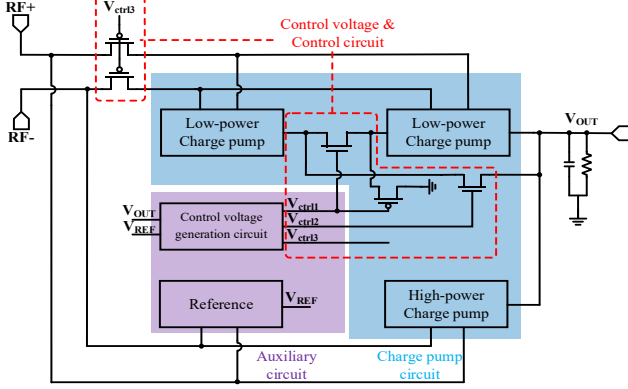


Fig. 4. Structure of the proposed charge pump system.

charge pump and the cross-coupled (CC) charge pump. As shown in Fig. 2(a), the conventional Dickson charge pump alternately transfers charges in the forward direction under the action of RF signals. When $RF_+ < RF_-$, M_{N1} and M_{N3} are turned on while M_{N2} is turned off. In this state, the input voltage V_{in} charges capacitor C_1 , and at the same time, capacitor C_2 supplies charges to the load capacitor. When $RF_+ > RF_-$, M_{N2} is turned on while the other transistors are turned off, and capacitor C_1 transfers charges to C_2 . Although the Dickson charge pump structure has the advantages of a simple circuit and a significant boost effect, its inherent defects are particularly prominent in low-power scenarios: When the input power (P_{IN}) and input voltage (V_{IN}) are low, due to the threshold voltage (V_{TH}) of MOS transistors, diode-connected MOS transistors in the circuit will be difficult to conduct, thus unable to generate sufficient forward current, resulting in extremely low PCE of the circuit [8].

Another classic CC charge pump is depicted in Fig. 2(b). When $RF_+ > RF_-$, the transistors M_{N2} and M_{P1} are turned on, while M_{N1} and M_{P2} are turned off. In this state, V_{in} charges capacitor C_2 and C_1 charges to the load capacitor. When $RF_+ < RF_-$, the switching states of the above four transistors are reversed and provides charging path for C_1 and load capacitor. Benefiting from its dynamic threshold voltage cancellation technology, the CC charge pump exhibits a relatively low on-resistance in low-power scenarios, thus achieving high transmission efficiency. However, under high P_{in} conditions, due to the small on-resistance of both PMOS and NMOS in the CC charge pump, the leakage current is large and PCE is relatively low [9]. From the above, both types of charge pumps have their own advantages and disadvantages under high and low P_{IN} conditions respectively, which also results in a single charge pump being unable to accommodate a wide range of P_{IN} . Therefore, how to design the charge pump structure to ensure the entire system achieves excellent conversion efficiency across a wide range of P_{IN} is the main challenge in current research [10]-[13].

In this paper, we propose a wide dynamic range charge pump featuring three mutually switchable operating modes, with an auxiliary circuit providing control signals for mode switching. This paper is organized as follows: Section II introduces and analyzes the operating principles of the proposed design, Section III presents the simulation results, and Section IV draws conclusions.

II. DESIGN OF THE PROPOSED CHARGE PUMP

Fig. 3 illustrates the conceptual structure of the proposed design. Traditional single-path charge pumps maintain high PCE only within a narrow P_{IN} range, whereas combining two paths effectively broadens PDR. Based on this dual-path integration, this work further develops a charge pump system with three switching modes, whose components and operating principles are elaborated below.

A. Structure and Components

Fig. 4 shows the structure diagram of the proposed design, which mainly consists of three charge pump circuits for high power and low power scenarios, a reference voltage generation circuit, a control voltage generation circuit, and a basic path selection control circuit. The low-power charge pump module is primarily based on a 2 stage CC charge pump, which exhibits superior PCE under low P_{IN} conditions. Conversely, under high P_{IN} scenarios, the system employs a traditional 2 stage Dickson charge pump with diode-connected MOS transistors. Owing to the absence of reverse leakage in diode-connected MOS transistors, this configuration effectively minimizes reverse current and enhances efficiency at high power levels.

Beyond the two charge pump units, the system incorporates a reference voltage generation circuit and a control voltage generation circuit. Both circuits function to produce control signals for the mode selection logic, enabling the system to switch between different operational modes across varying P_{IN} conditions. It is important to note that these auxiliary circuits consume a portion of the RF harvested energy; thus, their power consumption characteristics must be carefully considered during the design phase to prioritize low-power, high-performance implementations.

B. Operating principles

This section elaborates on the charge pump system's operating modes under different power conditions. As shown in Fig. 5(a), when the input power is low, the input voltage and current at the input terminal are relatively low. If only a second-order CC charge pump serves as the primary rectification component, its transmission efficiency is certainly high, but it fails to reach the desired voltage. This work employs a cascade of two CC charge pumps, where the output of one serves as the input to the other. On the one hand, this ensures the transmission efficiency of the low-power charge pump under low-power conditions; on the other hand, it increases the output voltage, thus preventing the load from malfunctioning due to insufficient output voltage under low-power conditions. At this point, the overall transmission efficiency of the charge pump is as follows:

$$PCE_{series} = \frac{P_{OUT_L}}{P_{IN_L}} * \frac{P_{OUT_L}}{P_{IN_L}} \quad (1)$$

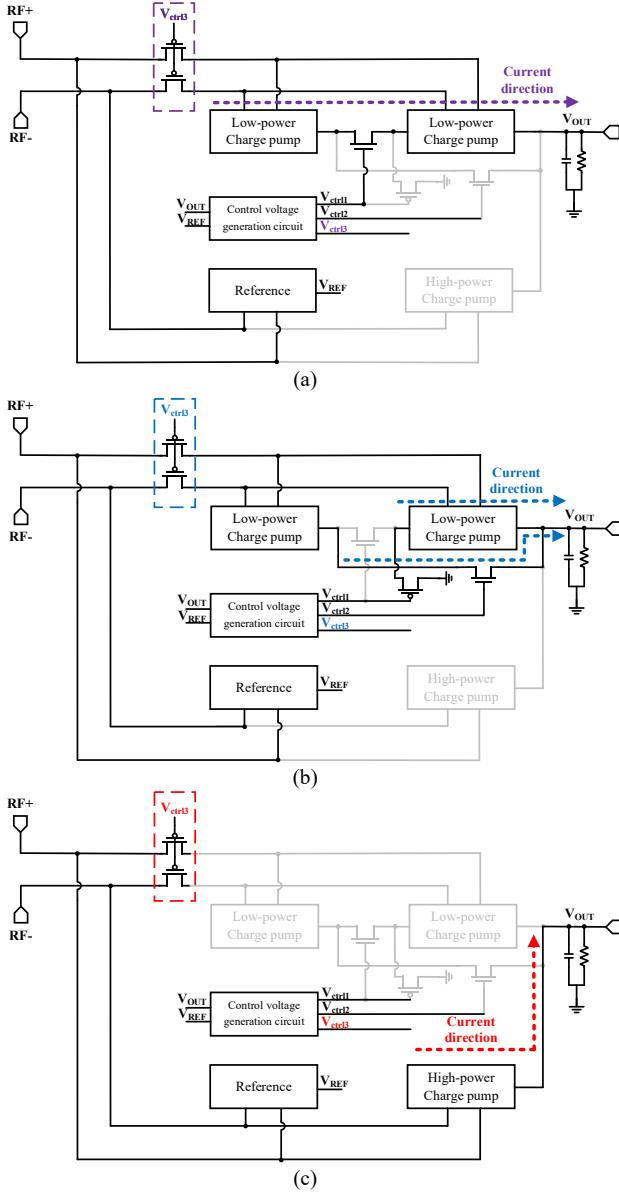


Fig. 5. System operation status in (a) low power condition, (b) medium power condition, (c) high power condition.

where P_{OUT_L} and P_{IN_L} is the output power and input power of the low-power charge pump.

When the input power increases to medium level, the two low-power charge pumps can be reconfigured from series to parallel via control signals from the control circuit, as shown in Fig. 5(b). At this point, the transmission efficiency expression of the charge pump system is as follows:

$$PCE_{parallel} = \frac{P_{OUT_L} + P_{OUT_L}}{P_{IN_L} + P_{IN_L}} = \frac{P_{OUT_L}}{P_{IN_L}} \quad (2)$$

As can also be inferred from the expression, since $0 < PCE < 1$, the square of this term is smaller than the term itself. This indicates that the transmission efficiency in the parallel mode is superior to that in the series mode.

When the system is working in the low-power and mid-power mode, the impact of high-power charge pump is neglectable. Firstly, the amplitude of the AC signal from the receiving terminal is relatively small at this point, which may not exceed the V_{th} of the diode-connected transistor, and thus cannot deliver power to the high-power charge pump. Secondly, the high-power charge pump exhibits higher impedance, the vast majority of the current naturally flows to the low-power charge pump, so the high-power charge pump has little impact on the overall PCE.

Under high-power conditions as shown in Fig. 5(c), the low-power charge pump exhibits large reverse current, which significantly degrades the overall transmission efficiency. The system's control signal disconnects the switches between the RF input terminal and the input of the low-power charge pump, thus preventing energy input to the low-power charge pump under high-power conditions. Additionally, with the high input power and voltage at this stage, the impact of the diode-connected transistor's threshold voltage is relatively minor. At this point, the entire system relies solely on the high-power charge pump for energy transfer; thus, the transmission efficiency of the entire system is as follows:

$$PCE = \frac{P_{OUT_H}}{P_{IN_H}} \quad (3)$$

where P_{OUT_H} and P_{IN_H} is the output power and input power of the high-power charge pump.

The control signal generation circuit in this work, consisting of a comparator, inverter, and diode, takes the charge pump system's output voltage V_{OUT} and a reference voltage as inputs to generate three control signals V_{ctrl1} , V_{ctrl2} , V_{ctrl3} for mode selection. Its temporal logic operates as follows: in low power scenario, V_{OUT} is low, driving the control signal generation circuit to output high V_{ctrl1} , low V_{ctrl2} and low V_{ctrl3} , activating series-connected switches to configure the two low-power charge pumps in series. As power increases, V_{OUT} rises, and V_{ctrl2} becomes high while V_{ctrl3} still stays low, transitioning the two low-power charge pumps to parallel connection. In high power condition, further increases in V_{OUT} maintain V_{ctrl1} low and V_{ctrl2} high, with V_{ctrl3} also turning high. This signal disconnects the RF input terminal from the low-power path, leaving only the high-power charge pump active, thus minimizing reverse current and enhancing transmission efficiency.

III. RESULTS AND COMPARISON

The proposed charge pump system is designed and implemented in a standard 0.18 μ m CMOS technology. Fig. 6 presents the PCE of each individual path and the overall simulated PCE of the proposed design. Simulation results indicate that each path maintains favorable PCE within its respective narrow input power range. Meanwhile, the adaptive mode selection of the proposed design significantly expands the dynamic range from -22dBm to 7dBm. Additionally, the design

TABLE I. PERFORMANCE COMPARISON WITH PREVIOUS WORKS

Parameter	TCASII 2017[14]	ISCAS 2022[15]	TCASI 2020[16]	TPE 2019[17]	JSSC 2019[18]	JSSC 2021[19]	This Work*
Tech(nm)	65	180	180	180	65	65	180
Freq(GHz)	0.9	2.45	2.4	0.9	2.45	2.45	2.45
Stage	5	2	3	N.A.	1	1	2/4
Structure	Dual-path	Dual-path	Dual-path	MPPT	MPPT	MPPT	Triple-path
Peak PCE @ P_{IN}	36.5% @-10dBm	62% @1.6dBm	47.1% @- 6dBm	48.2% @0dBm	48.3% @-3dBm	32.3% @-14.1dBm	61.4% @-10dBm
Dynamic Range**	11dB	27dB	16dB	40dB	N.A.	15dB****	29dB
Sensitivity @ V_{out} ***	-17.7dBm @1V	-20.7dBm @1V	-16dBm @1V	-20dBm @1V	-17.1dBm @400mV	-26.7dBm @400mV	-21dBm @1V
$R_L(k\Omega)$	147	50	5	23	MPPT PMU	MPPT PMU	100

*Simulation result, **PCE above 20%, *** $R_L=\infty$, ****Estimated from the figure.

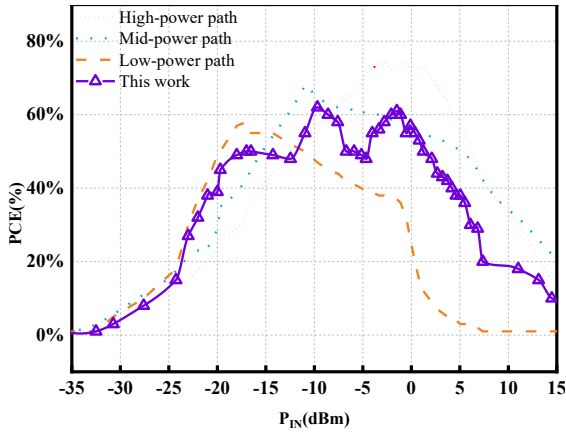


Fig. 6. Simulated PCE of the three individual mode and the proposed design with the three combination modes.

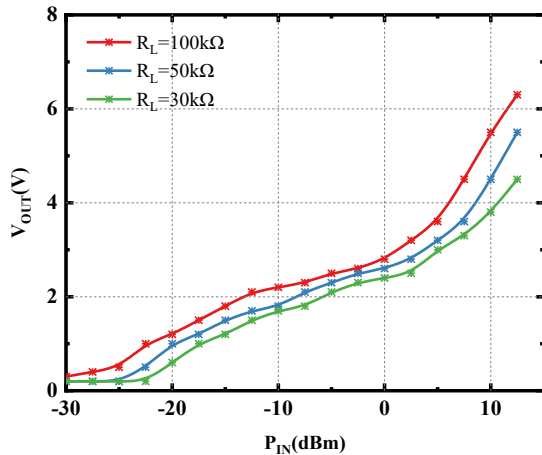


Fig. 7. Output voltage versus input power with different load.

achieves three peak PCE values across the three operating modes, with a maximum PCE of 61.4% observed at an input power of -10dBm.

Fig. 7 shows that the output voltage of the charge pump system versus the input power P_{IN} with different loads. The minimum P_{IN} required to charge up the charge pump with 1V output is -21dBm under pure capacitive load ($R_L=\infty$). Table I summarizes the performance results of the proposed charge pump system and compares with state-of-the-art published works. The three-mode switching structure designed in this work achieves a 29dB PDR, within which the PCE remains above 20%.

IV. CONCLUSION

This work presents a 2.45 GHz charge pump system tailored for RFEH, featuring an wide PDR and three switchable operating modes. To broaden the input power range where high PCE is sustained, two types of RF-DC rectifiers are deployed for low-power, mid-power and high-power paths, alongside a simple control circuit for mode selection. Implemented by a standard 0.18 μm CMOS process, the system demonstrates a sensitivity of -21 dBm and a peak PCE of 61.4% at -10 dBm. Under the conditions of 2.45 GHz operation and a 100 k Ω load, the proposed charge pump retains a PCE exceeding 20% across an input power range from -22 dBm to 7 dBm. This rectifier operates stably across a wider RF input power range, making it more suitable for power supply requirements of modern IoT devices.

REFERENCES

- [1] F. Mao, Y. Lu and R. P. Martins, "A Reconfigurable Cross-Connected Wireless-Power Transceiver for Bidirectional Device-to-Device Wireless Charging," in *IEEE Journal of Solid-State Circuits*, vol. 54, no. 9, pp. 2579-2589, Sept. 2019, doi: 10.1109/JSSC.2019.2921449.

- [2] A. Mahsifar and M. Yavari, "A High Dynamic Range Differential Rectifier for RF Energy Harvesting," 2021 29th Iranian Conference on Electrical Engineering (ICEE), Tehran, Iran, Islamic Republic of, 2021, pp. 124-128, doi: 10.1109/ICEE52715.2021.9544250.
- [3] A. C. C. Chun, H. Ramiah and S. Mekhilef, "Wide Power Dynamic Range CMOS RF-DC Rectifier for RF Energy Harvesting System: A Review," in IEEE Access, vol. 10, pp. 23948-23963, 2022, doi: 10.1109/ACCESS.2022.3155240.
- [4] D. Khan et al., "A High-Efficient Wireless Power Receiver for Hybrid Energy-Harvesting Sources," in IEEE Transactions on Power Electronics, vol. 36, no. 10, pp. 11148-11162, Oct. 2021, doi: 10.1109/TPEL.2021.3071374.
- [5] J. -R. Yan, H. -Y. Kuo and Y. -T. Liao, "A Wide-input-range 918MHz RF Energy Harvesting IC with Adaptive Load and Input Power Tracking Technique," ESSCIRC 2022- IEEE 48th European Solid State Circuits Conference (ESSCIRC), Milan, Italy, 2022, pp. 361-364.
- [6] Z. Wu, Y. Zhao, Y. Sun, H. Min and N. Yan, "A Self-Bias Rectifier with 27.6% PCE at -30dBm for RF Energy Harvesting," 2021 IEEE International Symposium on Circuits and Systems (ISCAS), Daegu, Korea, 2021, pp. 1-5.
- [7] C. Hegde, A. Mohan, S. Mondal and R. P. Paily, "A Wide Dynamic Range Differential Drive CMOS Rectifier for μ Watts RF Energy Harvesting Systems," 2025 38th International Conference on VLSI Design and 2024 23rd International Conference on Embedded Systems (VLSID), Bangalore, India, 2025, pp. 175-179, doi: 10.1109/VLSID64188.2025.00043.
- [8] T. Yim, C. Lee and H. Yoon, "A High Speed Modified Dickson Charge Pump," 2021 IEEE International Symposium on Circuits and Systems (ISCAS), Daegu, Korea, 2021, pp. 1-5.
- [9] Ming-Dou Ker, Shih-Lun Chen and Chia-Shen Tsai, "Design of charge pump circuit with consideration of gate-oxide reliability in low-voltage CMOS processes," in IEEE Journal of Solid-State Circuits, vol. 41, no. 5, pp. 1100-1107, May 2006, doi: 10.1109/JSSC.2006.872704.
- [10] Y. Tian and L. Liu, "A High-Sensitivity, Low-Power Dual-Band RF Energy Harvesting and Managing System for Bio-Potential Acquisition," 2021 IEEE 14th International Conference on ASIC (ASICON), Kunming, China, 2021, pp. 1-4.
- [11] W. W. Y. Lau and L. Siek, "2.45GHz wide input range CMOS rectifier for RF energy harvesting," 2017 IEEE Wireless Power Transfer Conference (WPTC), Taipei, Taiwan, 2017, pp. 1-4, doi: 10.1109/WPT.2017.7953896.
- [12] A. Ballo, A. D. Grasso and G. Palumbo, "A High-Performance Charge Pump Topology for Very-Low-Voltage Applications," in IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 67, no. 7, pp. 1304-1308, July 2020, doi: 10.1109/TCSII.2019.2932471.
- [13] A. F. Ponchet et al., "Analysis and Design of a Reconfigurable RF-DC Rectifier for RF Energy Harvesting Systems," 2024 IEEE Nordic Circuits and Systems Conference (NorCAS), Lund, Sweden, 2024, pp. 1-6, doi: 10.1109/NorCAS64408.2024.10752467.
- [14] Y. Lu et al., "A Wide Input Range Dual-Path CMOS Rectifier for RF Energy Harvesting," in IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 64, no. 2, pp. 166-170, Feb. 2017, doi: 10.1109/TCSII.2016.2554778.
- [15] X. Li, K. Wang, Y. Zhou and H. Zhang, "A 2.45 GHz Dual-Path CMOS RF-to-DC Rectifier with 27 dB Input Range and -20.7 dBm Sensitivity," 2022 IEEE International Symposium on Circuits and Systems (ISCAS), Austin, TX, USA, 2022, pp. 541-545.
- [16] S. Nagaveni, P. Kaddi, A. Khandekar and A. Dutta, "Resistance Compression Dual-Band Differential CMOS RF Energy Harvester Under Modulated Signal Excitation," in IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 67, no. 11, pp. 4053-4062, Nov. 2020, doi: 10.1109/TCSI.2020.3006156.
- [17] S. -Y. Kim et al., "A -20 to 30 dBm Input Power Range Wireless Power System With a MPPT-Based Reconfigurable 48% Efficient RF Energy Harvester and 82% Efficient A4WP Wireless Power Receiver With Open-Loop Delay Compensation," in IEEE Transactions on Power Electronics, vol. 34, no. 7, pp. 6803-6817, July 2019, doi: 10.1109/TPEL.2018.2872563.
- [18] P. Xu, D. Flandre and D. Bol, "Analysis, Modeling, and Design of a 2.45-GHz RF Energy Harvester for SWIPT IoT Smart Sensors," in IEEE Journal of Solid-State Circuits, vol. 54, no. 10, pp. 2717-2729, Oct. 2019, doi: 10.1109/JSSC.2019.2914581.
- [19] P. Xu, D. Flandre and D. Bol, "A Self-Gating RF Energy Harvester for Wireless Power Transfer With High-PAPR Incident Waveform," in IEEE Journal of Solid-State Circuits, vol. 56, no. 6, pp. 1816-1826, June 2021, doi: 10.1109/JSSC.2021.3061196.